

7. Static Malware Analysis

1. Definition

Static Malware Analysis is the process of analyzing a suspicious executable **without executing it**, by inspecting its **binary structure, metadata, and embedded artifacts**.

It is widely used in Digital Forensics to:

- Avoid system compromise
- Perform safe investigation
- Extract early threat intelligence

Unlike dynamic analysis, it does **not reveal runtime behavior**, but gives strong clues about intent.

2. Detailed Objectives

- Identify file type and architecture (PE32, PE64, ELF, script)
- Determine compiler/packer used
- Extract Indicators of Compromise (IOCs)
- Understand possible malware functionality
- Detect obfuscation or anti-analysis techniques
- Generate detection signatures (YARA)

3. Core Concepts

3.1 PE File Structure (Windows Executables)

A Portable Executable (PE) file is the standard format for .exe and .dll files.

PE Structure Breakdown:

- **DOS Header (MZ)** → Legacy compatibility
- **PE Header (Signature: PE\0\0)** → File metadata
- **Optional Header** → Entry point, image base
- **Section Table** → Maps sections



Important Sections:

Section	Purpose
.text	Executable code
.data	Initialized variables
.bss	Uninitialized data
.rsrc	Icons, images, payloads
.idata	Imported libraries

Suspicious signs:

- Missing or renamed sections
- Very high entropy
- Abnormal entry point

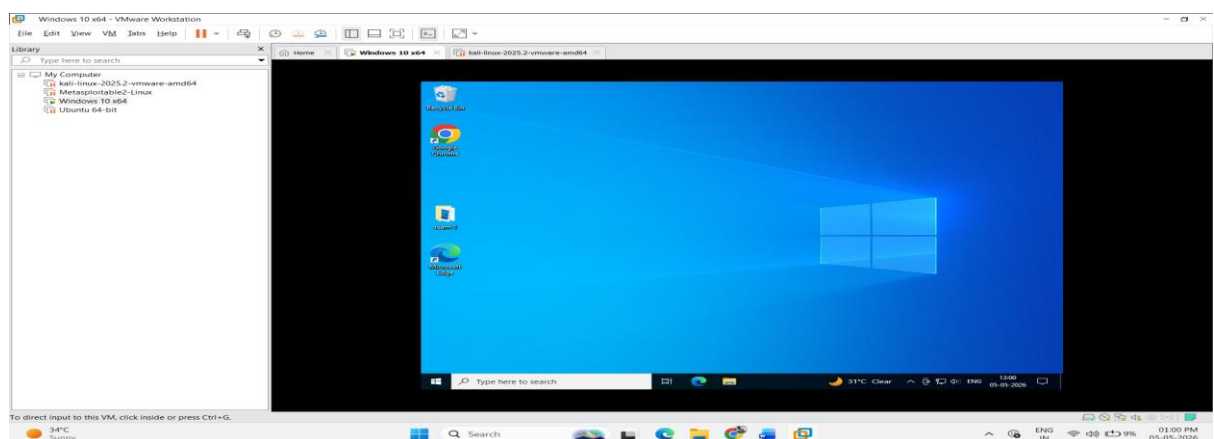
4. Lab Setup

Use:

- VirtualBox / VMware
- Snapshot before analysis

Best Practices:

- No internet OR controlled network
- Disable clipboard sharing
- Use non-production system



5. Practical

Step 1: File Identification & Metadata

Tools:

- file
- ExifTool

Commands:

file malware.exe
exiftool malware.exe

Analysis:

- File type → PE32 executable
- Compile time → (Fake timestamps possible)
- Author/company info

Red Flags:

- Timestamp mismatch (e.g., 1992 or future date)
 - Unknown compiler
-

Step 2: Cryptographic Hashing

Commands:

md5sum malware.exe
sha256sum malware.exe

Why important:

- Unique fingerprint of file
- Used for threat intelligence

Check on:

- VirusTotal

Interpretation:

- Detection ratio (e.g., 22/70 engines)



CYART

inquiry@cyart.io

www.cyart.io

- Malware family name
- Behavior tags

The screenshot shows the VirusTotal web interface for the domain `amazon-clone-eta-one.vercel.app`. The domain is flagged as malicious by 22 out of 93 security vendors. The analysis shows the following details:

- Community Score:** 22 / 93
- Registrar:** Tucows Domains Inc
- Creation Date:** 6 years ago
- Last Analysis Date:** 1 minute ago

The **DETECTION** tab is active, showing a table of security vendors' analysis results:

Security vendors' analysis	Do you want to automate checks?
ADMINUSLabs	Malicious
ArcSight Threat Intelligence	Malware
Criminal IP	Phishing
Emsisoft	Phishing
alphaMountain.ai	Phishing
BitDefender	Phishing
CyRadar	Phishing
Ermes	Phishing



Field	Value
Dataset Source	PhishTank / URLhaus
Sample / Domain	amazon-clone-eta-one.vercel.app
Category	Phishing
Malware Type	E-commerce Phishing / Fake Amazon Login Clone
Tool Used	VirusTotal
Detection	22/93 flagged malicious
Detection Details	Multiple vendors detected phishing, fake Amazon clone site
Score	22/93
Remark	Fake Amazon clone used for credential/payment harvesting
Field	Value
Dataset Source	PhishTank / URLhaus
Sample / Domain	amazon-clone-eta-one.vercel.app
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Score	22/93
Remark	Fake Amazon clone used for credential/payment harvesting

Step 3: String Analysis

Tools:



- strings
- FLOSS (for obfuscated malware)

Command:

strings malware.exe > output.txt

Extract:

- URLs → http://badsite.com
- IP → 192.168.x.x
- Registry keys → HKCU\Software\...
- File paths → C:\Windows\System32

This gives **direct insight into malware behavior**.

Sample: Lab01-01.dll

```
strings64 "Practical Malware Analysis Lab 01-01.dll_"
```

```
C:\Windows\system32\cmd.exe
No matching files were found.

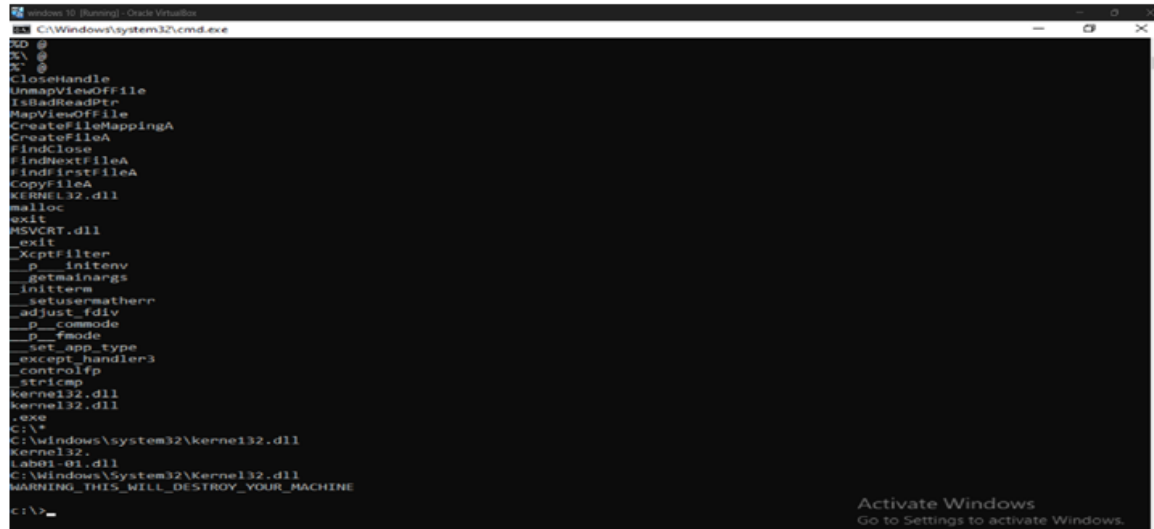
c:\>strings64 "C:\Users\win10\Downloads\capa-testfiles-master\capa-testfiles-master\Practical Malware Analysis Lab 01-01.dll_"

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

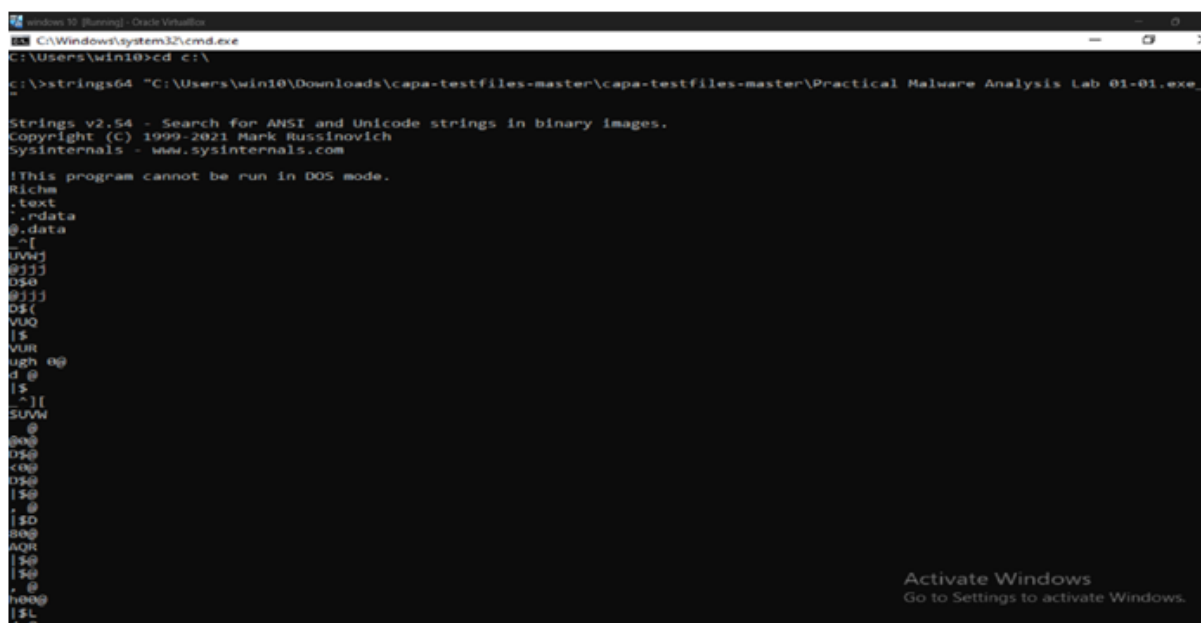
!This program cannot be run in DOS mode.
Rich
Text
.data
.data
.reloc
SUV
80'
80'
L$XQH
h('
RVF
05'
C(
10h '
[14
045
[14PQJ
05VD
[1
07h
2d'
V'3
LX'
WVS
WVS
WVS
07WPS
08WVS
WVS
*[]
CloseHandle
Sleep
```



```
strings04 "Practical Malware Analysis Lab 01-01.exe_"
```



```
C:\Windows\system32\cmd.exe
C:\>strings04 "Practical Malware Analysis Lab 01-01.exe_"
C:\>
C:\>
C:\>
CloseHandle
UnmapViewOfFile
IsBadReadPtr
MapViewOfFile
CreateFileMappingA
CreateFileA
FindClose
FindNextFileA
FindFirstFileA
CopyFileA
kernel32.dll
malloc
exit
MSVCRT.dll
_exit
XcptFilter
_p__initenv
_getmainargs
_initterm
_setusermatherr
_adjust_fdiv
_p__commode
_p__fmode
_set_app_type
_except_handler3
_controlfp
_stricmp
kernel32.dll
kernel32.dll
.exe
C:\*
C:\windows\system32\kernel32.dll
kernel32.
lab01-01.dll
C:\Windows\System32\Kernel32.dll
WARNING_THIS_WILL_DESTROY_YOUR_MACHINE
C:\>
```



```
C:\Windows\system32\cmd.exe
C:\Users\win10>cd c:\
C:\>strings64 "C:\Users\win10\Downloads\capa-testfiles-master\capa-testfiles-master\Practical Malware Analysis Lab 01-01.exe_"
Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (c) 1999-2021 Mark Russinovich
sysinternals - www.sysinternals.com

!This program cannot be run in DOS mode.
Richm
text
.rdata
.data
.^[
UVWj
0j33
030
0j1j
03(
VUQ
j$
VUR
ugn 0@
d 0
j$
.^][
SUVW
0
00@
03@
00@
03@
j$@
0
j$0
00@
AQR
j$@
j$@
0
000@
j$L
d 0
```

Step 4: PE Header Analysis

Tools:

- PView
- CFF Explorer
- objdump

What to Analyze:

Entry Point:



- Where execution starts
- If inside unusual section → suspicious

Imports:

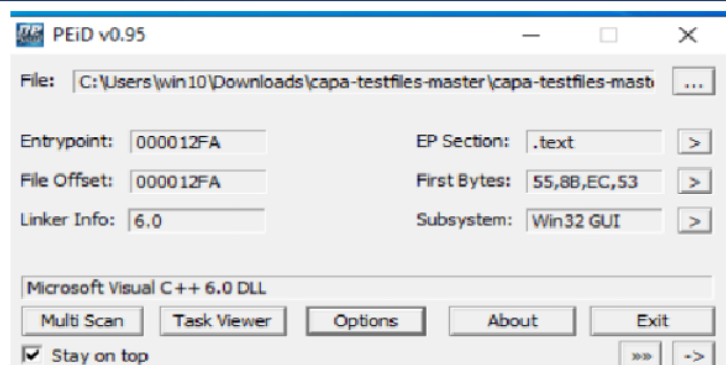
- kernel32.dll, user32.dll, ws2_32.dll

Suspicious APIs:

API	Meaning
CreateRemoteThread	Process injection
WriteProcessMemory	Code injection
WinExec	Command execution
InternetOpen	Network activity

Sample 1: Practical Malware Analysis Lab 01-01.dll

Detection	Microsoft Visual C++ 6.0 DLL
Entry Point	0x000012FA
EP Section	.text (Normal — not packed)
First Bytes	55 8B EC 53 (Standard x86 function prologue)
Subsystem	Win32 GUI
Linker Version	6.0
Packing Status	NOT PACKED — Direct static analysis is possible



Step 5: Detect Packing & Obfuscation

Tool:



- PEiD

Common Packers:

- UPX

```
C:\Windows\system32\cmd.exe
Ultimate Packer for executables
Copyright (C) 1996 - 2020
Jan 7th 2020
UPX 5.1.0 Markus Oberhumer, Laszlo Molnar & John Reiser
upx: Practical Malware Analysis Lab 01-01.dll_: FileNotFoundException: Practical Malware Analysis Lab 01-01.dll_: No such file or directory
Tested 0 files.
C:\Users\win10\Downloads\upx-5.1.0-win64\upx-5.1.0-win64\upx -t "Practical Malware Analysis Lab 01-01.exe_"
Ultimate Packer for executables
Copyright (C) 1996 - 2020
Jan 7th 2020
UPX 5.1.0 Markus Oberhumer, Laszlo Molnar & John Reiser
upx: Practical Malware Analysis Lab 01-01.exe_: FileNotFoundException: Practical Malware Analysis Lab 01-01.exe_: No such file or directory
Tested 0 files.
C:\Users\win10\Downloads\upx-5.1.0-win64\upx-5.1.0-win64\upx -t "Practical Malware Analysis Lab 01-02.exe_"
Ultimate Packer for executables
Copyright (C) 1996 - 2020
Jan 7th 2020
UPX 5.1.0 Markus Oberhumer, Laszlo Molnar & John Reiser
upx: Practical Malware Analysis Lab 01-02.exe_: FileNotFoundException: Practical Malware Analysis Lab 01-02.exe_: No such file or directory
Tested 0 files.
C:\Users\win10\Downloads\upx-5.1.0-win64\upx-5.1.0-win64\upx -t "Practical Malware Analysis Lab 01-04.exe_"
Ultimate Packer for executables
Copyright (C) 1996 - 2020
Jan 7th 2020
UPX 5.1.0 Markus Oberhumer, Laszlo Molnar & John Reiser
upx: Practical Malware Analysis Lab 01-04.exe_: FileNotFoundException: Practical Malware Analysis Lab 01-04.exe_: No such file or directory
Tested 0 files.
C:\Users\win10\Downloads\upx-5.1.0-win64\upx-5.1.0-win64\upx -t "Practical Malware Analysis Lab 01-02.dll_"
Ultimate Packer for executables
Copyright (C) 1996 - 2020
Jan 7th 2020
UPX 5.1.0 Markus Oberhumer, Laszlo Molnar & John Reiser
upx: Practical Malware Analysis Lab 01-02.dll_: FileNotFoundException: Practical Malware Analysis Lab 01-02.dll_: No such file or directory
Tested 0 files.
C:\Users\win10\Downloads\upx-5.1.0-win64\upx-5.1.0-win64\upx -t "Practical Malware Analysis Lab 01-02.dll_"
Ultimate Packer for executables
Copyright (C) 1996 - 2020
Jan 7th 2020
UPX 5.1.0 Markus Oberhumer, Laszlo Molnar & John Reiser
upx: Practical Malware Analysis Lab 01-02.dll_: FileNotFoundException: Practical Malware Analysis Lab 01-02.dll_: No such file or directory
Tested 0 files.
```

- Themida
- ASPack

Indicators:

- High entropy (>7.0)
- Single section file
- No readable strings

Packed malware hides real code → must unpack later.

Step 6: Resource Analysis

Tools:

- Resource Hacker
- PE Explorer

Check:

- Embedded executables
- Images/icons



- Scripts

Malware often hides payload in .rsrc

Step 7: Advanced Static Techniques

Disassembly (Reverse Engineering)

Tools:

- IDA Free
- Ghidra

Convert binary → assembly code

Indicators:

- Hardcoded encryption keys
 - Function logic
 - Control flow
-

Step 8: YARA Rule Creation (Detection)

Purpose:

Detect malware automatically based on patterns

Example:

```
rule Banking_Trojan
{
  meta:
    author = "Analyst"
    description = "Detects banking malware"
  strings:
    $url = "login-bank.com"
    $api = "CreateRemoteThread"
  condition:
    $url and $api
}
```

6. Indicators of Compromise (IOC Extraction)

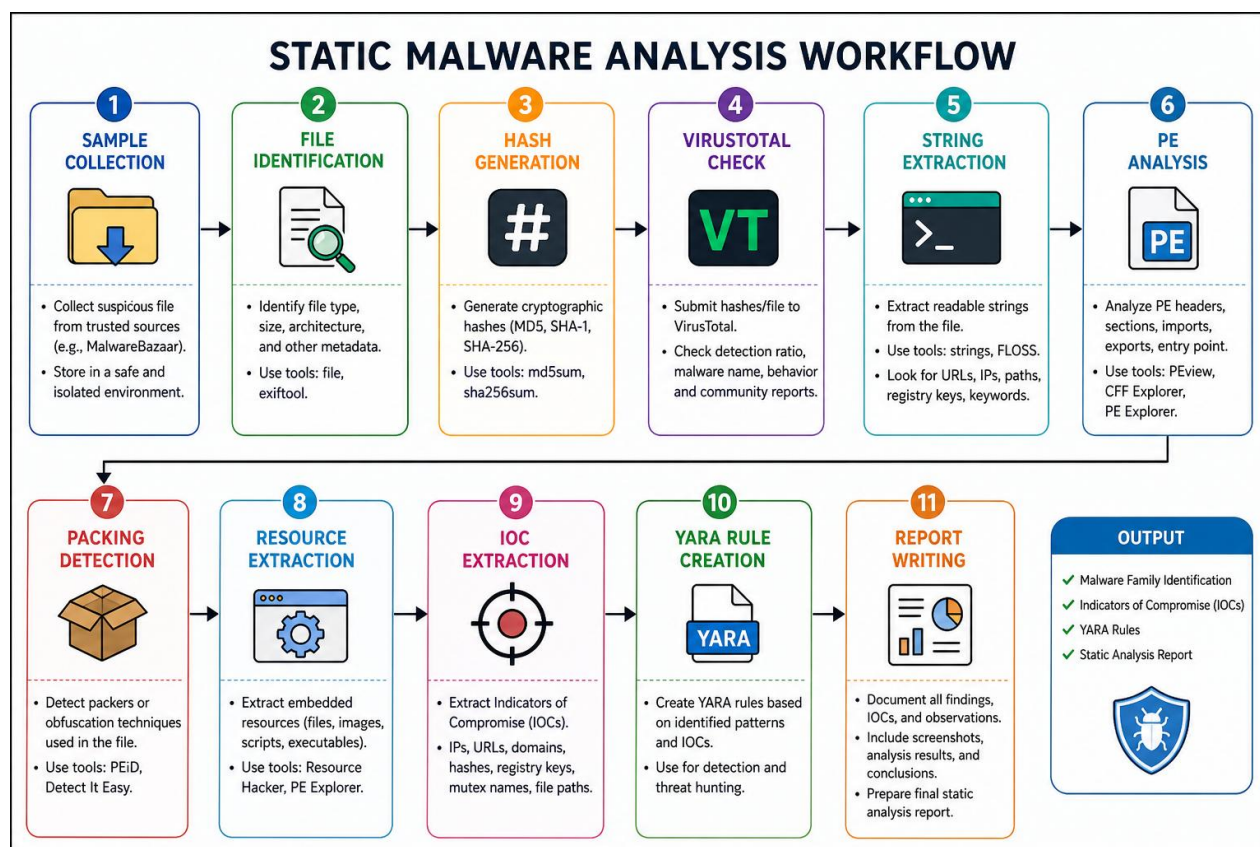
Types:

- File Hashes
- IP Addresses
- Domains
- Registry Keys
- Mutex names

These are used in:

- SIEM tools
- Threat hunting
- Incident response

7. Complete Workflow





8. Limitations of Static Analysis

- Cannot detect runtime behavior
- Packed/encrypted malware difficult
- Anti-analysis techniques (junk code, obfuscation)

That's why dynamic analysis is also required.

9. Key Exam Points

- Static = No execution
- Safe but limited
- PE structure is MUST
- Strings = quickest insight
- Hash = identification
- YARA = detection

10. Conclusion

Static Malware Analysis is a foundational step in malware investigation that enables analysts to safely extract critical information such as file structure, embedded artifacts, and IOCs. While it has limitations in detecting runtime behavior, it plays a crucial role in early-stage threat detection, classification, and preparation for deeper dynamic analysis.